CarbonSense Intelligence

An ML-powered carbon operations platform that predicts energy demand, estimates emissions and recommends when flexible loads should move.

FORECAST WINDOW	MODEL MAE	INTERVAL COVERAGE	ACTION ENGINE
1-168h	**2.41**	**89.72%**	**Carbon Shift**
hourly outlook	kWh on demo test	target: 90%	low-carbon scheduling

01 / WHY IT EXISTS

From energy data to an operating decision

Most carbon dashboards explain what happened after the energy was consumed. CarbonSense looks ahead, quantifies uncertainty and turns the forecast into a scheduling action.

01

Reactive reporting

Carbon totals arrive too late to change the operating schedule.

02

Hidden uncertainty

Single-number forecasts can create false confidence for operators.

03

No action layer

Charts often stop before recommending what load should move and when.

THE CARBONSENSE RESPONSE

- Forecast demand for the next 1 to 168 hours from time, lag, weather and renewable drivers.
- Keep emissions conversion outside the ML model so regional factors can change independently.
- Expose a 90% uncertainty range, peak demand, clean-energy coverage and data-quality evidence.
- Recommend the cleanest operating window and estimate savings from shifting flexible load.

02 / HOW IT WORKS

A reproducible time-series pipeline

The design separates demand forecasting, emissions accounting and decision logic.

01	02	03	04	05
Hourly data	**Features**	**ML forecast**	**CO2 service**	**Carbon Shift**
Energy, weather, solar and wind	Seasonality, lags and rolling statistics	Regularized demand regression	Regional or custom grid factor	Clean-window recommendation

MODEL FEATURES

- Cyclical hour features and day-of-week context
- 1-hour, 24-hour and 168-hour demand lags
- 24-hour rolling mean and standard deviation
- Temperature, solar generation and wind generation
- Recursive future prediction without random shuffling

TRUST CONTROLS

- Chronological train, validation and untouched test windows
- Previous-day seasonal baseline for honest comparison
- Validation-calibrated residual quantiles
- Wider uncertainty for longer horizons
- Versioned model and preprocessing artifact

CARBON CALCULATION

$$\text{predicted CO2 kg} = \text{predicted energy kWh} \times \text{carbon intensity}$$

The energy model never learns a fixed emissions factor. A verified provider can replace the demo factor without retraining demand prediction.

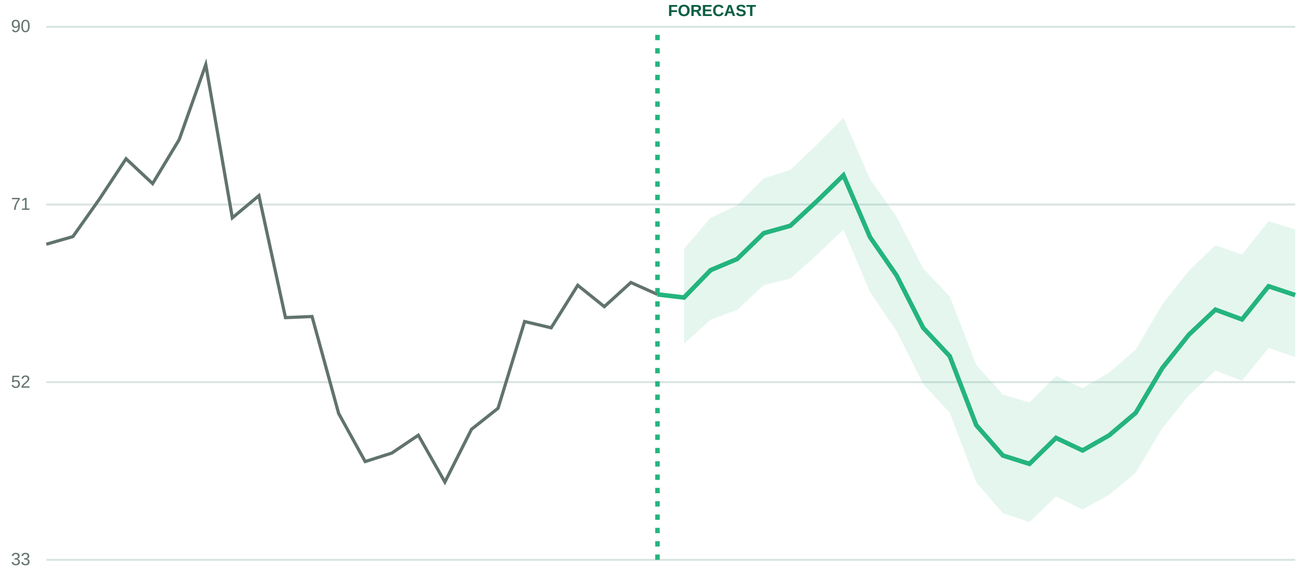
03 / WHAT IT PRODUCES

Forecasts designed for decisions

FORECAST ENERGY	FORECAST EMISSIONS	AVG INTENSITY	PEAK DEMAND
1392	**606.7**	**0.436**	**74.1**
kWh / next 24 hours	kgCO2e / next 24 hours	kgCO2e per kWh	kWh at peak hour

Actual demand to forecast

kWh - 90% forecast range



CARBON SHIFT - THE UNIQUE ACTION LAYER

Shift 7.4 kWh to the cleanest forecast hour

0.4 kgCO2e

estimated avoided emissions for this 24-hour scenario

The dashboard also exposes hourly CO2, renewable share, 90% ranges, the cleanest three-hour window and a downloadable CSV for operations teams.

04 / WHERE IT HELPS

One engine, multiple operating contexts

1 BUILDINGS

Forecast HVAC and office demand; shift charging or cooling to cleaner hours.

2 FACTORIES

Move flexible production, reduce peaks and test efficiency investments.

3 CAMPUSES

Coordinate solar, EV charging and multi-building sustainability targets.

4 MICROGRIDS

Align loads with renewables and supply demand forecasts to GridOpt.

WHY IT STANDS OUT

Actionable

Moves from prediction to a quantified operating recommendation.

Scenario-ready

Tests efficiency, renewables, temperature and grid intensity instantly.

Explainable

Shows inputs, baseline, uncertainty, data quality and model evidence.

05 / DEVELOPER HANDOFF

A small, portable technology stack

DASHBOARD	HTML5 + CSS3 + JavaScript + responsive SVG
REST API	FastAPI + Pydantic + OpenAPI + CORS
ML PIPELINE	Python + NumPy + Pandas + time-series features
DELIVERY	Pytest + Uvicorn + Docker + health checks

CORE ENDPOINTS

GET	`/api/v1/energy/forecast`	Demo and scenario forecast
POST	`/api/v1/energy/forecast/custom`	Real-history forecast
GET	`/api/v1/energy/history`	Actual chart series
GET	`/api/v1/data/quality`	Completeness and freshness
GET	`/api/v1/model/metrics`	Evaluation evidence

06 / VALIDATION AND NEXT STEPS

Strong evidence, clear production boundary

MODEL MAE	BASELINE MAE	MODEL MAPE	90% COVERAGE
2.41	4.17	3.93%	89.72%
kWh	kWh	safe denominator	held-out test

CHRONOLOGICAL EVALUATION

TRAIN + VALIDATION 1593 rows | TEST 399 rows

Time is never randomly shuffled. The earlier observations fit and calibrate the pipeline; the latest 399 observations remain untouched until final evaluation.

PRODUCTION ROADMAP

- Connect real smart-meter history and trusted future weather and renewable feeds.
- Replace illustrative factors with a verified regional or marginal-emissions provider.
- Add authentication, tenant isolation, rate limits, lineage and artifact storage.
- Monitor forecast error, interval coverage, missing data and feature drift.

HACKATHON PITCH

CarbonSense does not just predict electricity demand. It predicts when demand will occur, estimates the carbon impact and recommends when flexible loads should move.

Result: a measurable, explainable path from raw hourly data to lower-carbon operations.